# Saptara

9 November 2026

Arithmetic Progressions

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1. An arithmetic progression has $2n+1$ terms. Prove that the ratio of the sum of the odd-positioned terms to the sum of the even-positioned terms is $(n+1) : n$.

2. The sum of the first $p$ terms of an A.P. is $q$, and the sum of the first $q$ terms is $p$, where $p > q$. Show that the sum of the first $p+q$ terms is $-(p+q)$. Also find the sum of the first $p-q$ terms.

3. The first term of an A.P. is 2, and the sum of the first four terms equals one-fourth of the sum of the next five terms. Find the sum of the first 30 terms.

4. For each ratio of the sums of the first $n$ terms of two arithmetic series, find the requested ratio of terms.

   (i) $(14-4n) : (3n+5)$. Find the ratio of their 8th terms.

   (ii) $(2+3n) : (3+2n)$. Find the ratio of their 7th terms.

5. (i) If the sums of the first $m$ and $n$ terms of an A.P. are in the ratio $m^2 : n^2$, show that its $p$th and $q$th terms are in the ratio $(2p-1) : (2q-1)$.

   (ii) The sums of the first $m$ and $n$ terms of an A.P. are in the ratio $(2m+1) : (3n+1)$. Find the ratio of its 7th and 10th terms.

6. Let $S_k$ denote the sum of the first $k$ terms of an A.P.

   (i) If $S_1 = 6$ and $S_7 = 105$, prove that

   $$S_n : S_{n-3} = (n+3) : (n-3).$$

   (ii) If $S_3 = 6$ and $S_6 = 3$, prove that

   $$2(2n+1)S_{n+4} = (n+4)S_{2n+1}.$$

7. The sum of the first six terms of an A.P. is 42. The ratio of the 10th term to the 30th term is $\frac{1}{3}$. Find the first and thirteenth terms.